A Multi-Axis Empirical Audit of Machine-Learning Intrusion Detection

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Abstract

Processed intrusion-detection benchmarks support controlled model comparison, but they do not establish capability across time, feature perturbations, datasets, rare-event prevalence, deployment constraints, or attacks withheld from development. We present a multi-axis empirical audit that evaluates these validity questions separately while preserving native metrics, frozen operating rules, and claim-to-artifact provenance. On one shared forward target, mean PR-AUC across five registered seeds was 0.2599 after random-natural development and 0.6388 after chronological-natural development; the ordering held in all five realizations, although frozen thresholds transferred poorly and controls did not identify chronology as the sole mechanism. The processed CSE-CIC-IDS2018 reference condition provides a contrast: the selected balanced tree ensemble achieved F1 0.9285 and PR-AUC 0.9776. Cross-dataset ranking was directional, with IDS2018-to-CICIDS2017 PR-AUC 0.667483 and reciprocal PR-AUC 0.108176. At 0.1% assumed attack prevalence, one high-PPV projection required 33.5 analyst-hours per day, while a low-workload projection reflected negligible detection yield. Not every result deteriorated under stronger validation: several eligible attack families retained stable discrimination, whereas other outcomes depended on learner or support. Stronger validation instead narrowed the claims that the evidence could support. The conclusions remain conditional on two related benchmark families, frozen feature bridges and realizations, analytic operational assumptions, eligible attack families, and one measured hardware environment.

Keywords: intrusion detection, machine learning, empirical evaluation, dataset shift, reproducibility

1. Introduction

Machine-learning intrusion detection is commonly summarized by predictive metrics on a benchmark test set [1, 2]. These metrics answer a useful but bounded question: under a specified representation, split, learner, and operating rule, how well does a fitted system distinguish the benchmark labels? Inferring that the same capability persists in later traffic, another collection pipeline, a rare-event population, or an attack family absent during development requires evidence under those conditions [3]. Benchmark performance and validated capability are therefore related, not interchangeable. The gap arises through several distinct mechanisms. Random partitions can mix temporal regimes or attack-family composition that forward evaluation separates [4, 3]. Dataset or extraction artifacts can support discrimination without transporting to a new domain [5]. Cross-dataset comparisons require defensible shared semantics rather than matching column names [6, 7]. Ranking metrics describe score ordering, whereas precision, recall, and false-positive rate describe a selected operating point [8]. At low attack prevalence, that point can yield sharply different precision and analyst workload [9, 10]. Computational feasibility is specific to the measured inference path and hardware [11], and attacks withheld from training remain conditional on the family, support, threshold, and unknown-class protocol [12, 13]. These problems should not be collapsed into a single degradation or robustness score. Temporal transfer, shortcut sensitivity, cross-dataset portability, operational utility, computational feasibility, and unseen-family behavior use different populations and native quantities. A ranking can remain informative while its frozen threshold fails; transfer can be substantial in one dataset direction and near baseline in the reverse; and high projected positive predictive value (PPV) can coincide with excessive workload. A connected evaluation must preserve such tensions rather than edit them into one success or failure narrative. We reconstruct that evaluation around CSE-CIC-IDS2018 and CICIDS2017. A conventional processed-data reference is followed by source-faithful temporal contrasts, controlled shortcut-feature perturbations, bidirectional semantic bridges, prior-shift and workload projections, hardware-specific profiling, leave-one-attack-family-out (LOAO) evaluation, and frozen seed and control analyses. Target-facing decisions were locked before their corresponding evaluations. Claims are reported in their native metrics and linked to populations, source artifacts, configurations, code, and provenance records. The evidence is heterogeneous. The reference result is strong, and

chronological development retains substantially better forward ranking than random development across the evaluated seeds, but transferred operating thresholds remain poor. The forward cross-dataset bridge retains ranking signal while the reverse bridge does not. Shortcut interventions show split interactions, yet matched placebos prevent a causal leakage claim. Several withheld families remain discriminable, whereas others are learner-dependent, ineligible, or weakly supported. High projected PPV and manageable analyst workload also fail to align uniformly. These counterexamples identify which inference changes under each validity condition. This study makes four contributions:

1. It applies an evidence-governed multi-axis evaluation design to machine-learning IDS, treating temporal validity, operating-point transfer, shortcut sensitivity, bidirectional cross-dataset transfer, prevalence stress, deployment profiling, held-out-family behavior, and realization stability as separate questions with native metrics.
2. It provides a claim-to-provenance layer that maps manuscript statements and numerical values to frozen evidence, protocols, artifacts, source material, and explicitly classified reproducibility limits, without representing static verification as a complete scientific rerun.
3. It documents a heterogeneous empirical pattern in which benchmark discrimination, forward ranking, transfer, operating-point behavior, operational projections, computational profiles, and held-out-family results survive or fail under different conditions; no single degradation or robustness score represents these outcomes.
4. It distills the evaluated evidence boundaries into an eight-item checklist linking each question to a native metric, a control, and an interpretation ceiling; the checklist is a study-derived reporting aid, not a claim of priority over existing guidance.

The contribution is an evaluation discipline, not a new IDS architecture or a universal verdict on machine-learning detection. The study asks which benchmark capabilities survived each evaluated validity condition, which became conditional, and what remained unresolved.

2. Related Work

2.1. 2.1 Benchmark and temporal validity

CICIDS-family releases provide documented, flow-derived data for machine-learning IDS evaluation [14, 15, 16]. Their usefulness does not make a

processed or rebalanced sample representative of a deployment population: performance remains conditional on construction, representation, class balance, and evaluation protocol [1, 17, 2]. The present study therefore retains its processed CSE-CIC-IDS2018 result as a reference condition and evaluates other validity questions on separately governed populations. Random partitioning can also obscure collection artifacts, near-duplicate structure, and temporal or family dependencies [5, 3]. Time-consistent security evaluation better matches a future-use direction but can simultaneously alter prevalence and class composition [4]. A temporal contrast is consequently evidence about the chosen validation geometry, not by itself proof of concept drift.

2.2. 2.2 Shortcut and explanation validity

Dataset identifiers, flawed extraction fields, and collection artifacts can be predictive without encoding transportable attack behavior [5, 3]. Removing a feature establishes sensitivity to that intervention; it does not establish leakage or a causal explanation for transfer failure. Placebos and composition controls are therefore essential to the interpretation here. Explanation methods add a separate validity question. LIME fits a local surrogate [18], while SHAP defines additive feature attributions and TreeSHAP supplies tree-specific algorithms [19, 20]. Stability, decision agreement, and fidelity are different properties [21]. The manuscript retains that distinction rather than using repeatability as a proxy for faithful local reconstruction.

2.3. 2.3 Cross-domain and operational evaluation

Cross-dataset studies show that IDS performance depends on dataset pairing, shared representation, and source-to-target direction [6, 7, 22]. Elangovan et al. extend this line with temporal and semantic generalization across four public datasets, including forward and cross-domain comparisons, but do not reproduce this audit's shortcut/placebo, prevalence/workload, hardware, or eligibility-gated LOAO chain [23]. Ha Thanh's TAN-IDS likewise frames transfer-aware, deployment-oriented evaluation around a unified NetFlow representation and adaptation scenarios; its target adaptation and deployment scope differ from this study's source-only frozen bridges and analytic workload projections [24]. Mishra et al. study harmonized cross-dataset multi-model evaluation with statistical testing and reproducibility logging, using different datasets, harmonization procedures, and estimands; the associated correction concerns the funding statement, not the reported results or methodology [25, 26]. Berhili et al. directly overlap with this study's protocol-governance

concerns: their available article excerpts address validation-split bias, seed sensitivity, preprocessing leakage, semantically aligned cross-dataset transfer, and deployable-versus-oracle calibration [27]. The accessible evidence also supports overlap on operating-point validity and transfer, while sections not independently reviewed here are not used to claim that the study lacks any method. The present audit differs in connecting these questions with separate temporal, shortcut/placebo, prevalence/workload, hardware, eligibility-gated LOAO, and claim-provenance boundaries rather than treating them as one estimand. Operational interpretation is constrained by the base-rate problem: sensitivity and specificity do not determine positive predictive value or false-alert burden when attacks are rare [9]. Prior-probability-shift analysis further assumes stable class-conditional behavior [28], while controlled alarm studies show that false-alarm rate can affect analyst precision and time on task [10]. The present workload and cost quantities are therefore scenario projections, not measured field outcomes.

2.4. 2.4 Computational, unknown-family, and reproducibility scope

Computational feasibility depends on the inference path, batch, software, and hardware [11]; constrained-device IDS demonstrations also show why feature-extraction and model-execution boundaries must be explicit [29]. Open-set recognition formalizes test-time unknown classes [30], and IDS studies have evaluated attacks withheld from development under study-specific protocols [12, 13]. Such results depend on eligibility, support, metric, and threshold and do not establish arbitrary future-attack detection. Reproducibility programs emphasize code, experimental detail, and independent regeneration [31], while adaptive-analysis results show that repeated holdout feedback can weaken ordinary guarantees [32]. Our contribution is not priority over these practices; it is their study-specific integration through frozen protocols, target-opening records, equivalence receipts, and claim-level provenance, with unresolved rerun gaps kept explicit.

3. Datasets and Provenance

3.1. 3.1 CSE-CIC-IDS2018 roles

CSE-CIC-IDS2018 serves two non-equivalent roles [15, 14]. The conventional reference uses the original processed, rebalanced binary table, with frozen memberships of 192,593 training, 48,149 validation, and 60,186 holdout rows. It supports controlled model and operating-point selection but is not a

natural-prevalence, session-independent, or forward population. Recovered temporal provenance showed that the inherited membership was unsafe for temporal-window evaluation. Temporal analyses therefore use source-faithful dated exports and separately governed memberships. Here, source-faithful means that authenticated columns, cleaning rules, and temporal roles were recovered from the research lineage; it does not imply raw-packet availability or complete reconstruction of every historical environment. The shared forward target preserves collection order and is scientifically distinct from the processed holdout. High reference discrimination is not relabeled as temporal evidence, and the source-faithful analysis does not make the processed reference naturalistic.

3.2. 3.2 CICIDS2017 and semantic bridges

CICIDS2017 supplies a second benchmark domain, selected packet/flow provenance, and the family/day chronology used for LOAO evaluation [16, 14]. The primary forward bridge trains on IDS2018 and evaluates 2,830,743 effective CICIDS2017 rows; the reverse bridge trains under the frozen CICIDS2017 source protocol and evaluates the IDS2018 February 28 target. The directions are not averaged because their learners, prevalence, representations, and target roles differ. Cross-dataset comparability follows frozen semantic contracts. Bridge62 contains the defensible shared predictors. Bridge70 adds aggregate-flag fields and retains both the published serialization and a preregistered correction. The correction changes an identified representation dependency; it is not selected after target performance. Two planned `GROUNDED_S4` cells were cancelled before target opening because exact durable physical-row membership could not be reconstructed without a new heuristic. They were neither approximated nor replaced.

3.3. 3.3 Attack-family populations

Known categories in the processed reference holdout describe error concentration, not novelty. LOAO evaluation instead requires zero exposure to the target family in training and validation, a day-atomic `TRAIN < VALIDATION < TARGET` ordering, and sufficient target support. DDoS, Port Scan, Web Attack, Bot, and Infiltration were eligible. DOS and AUTH_BRUTE_FORCE were structurally ineligible, while Infiltration remained descriptive because its target contained 36 positives. The primary family target pairs the withheld family with temporally matched benign traffic; a context target that includes known attacks remains secondary.

3.4. 3.4 Provenance and target governance

Provenance labels retain narrow meanings. `Raw-exact` denotes preserved source bytes; `published` denotes released artifact semantics; `corrected` denotes a frozen documented correction; `source-faithful` denotes a derived representation following authenticated semantics and locked cleaning; and `reconstructed` denotes recovery from surviving lineage evidence without claiming byte identity. Detailed parser investigations, transport receipts, storage events, and failed mechanism searches are routed to the supplement or repository unless needed to interpret a main result. Development data support fitting, validation selection, and protocol checks only within declared roles. Target-facing model scope, features, thresholds, and cleaning were frozen before the corresponding target result. No target-guided fitting, feature search, calibration, or threshold reselection occurred. Some targets were historically opened and later reused for preregistered stability analysis; they are not presented as fresh blind holdouts. Current validation-safe interfaces verify identities, schemas, values, and receipts without opening scientific targets or deserializing models.

4. Validation Framework and Methods

4.1. 4.1 Reference evaluation

The reference question is conventional within-table discrimination on the processed CSE-CIC-IDS2018 binary data. Stratified training, validation, and holdout memberships are frozen and disjoint. Scaling is fitted on training data only for historically scaled learners; tree boosting retains its recorded preprocessing. Validation narrows the baseline inventory to XGBoost and LightGBM and selects a balanced XGBoost threshold of 0.51 and a recall-oriented LightGBM threshold of 0.26. The holdout cannot revise either decision. ROC-AUC and PR-AUC report ranking; F1, F2, precision, recall, false-positive rate (FPR), and error counts report operating behavior. Historical paired, class-stratified analyses describe conditional uncertainty on the same holdout. This axis does not estimate natural prevalence, temporal validity, domain transfer, or deployment utility.

4.2. 4.2 Temporal validation

Temporal validation asks whether development geometry changes ranking and threshold transfer on one common forward IDS2018 target. Four cells

cross random or chronological membership with natural or training-only rebalanced prevalence. The target contains 1,374,133 rows: 375,345 attack and 998,788 benign. The tree-ensemble recipe, cleaning, memberships, and operating points are frozen; rebalancing applies only to training. PR-AUC and ROC-AUC describe ranking, while frozen-threshold precision, recall, F1, and FPR describe operating transfer. The stability layer repeats the designated random-natural and chronological-natural cells over five frozen seeds and reports means, standard deviations, and directional counts without best-seed selection. Family-aware controls test whether chronology alone explains the contrast. The result is conditional on the recovered chronology and finite seed set; it is not a significance test, a session-independence guarantee, or a causal estimate of drift.

4.3. 4.3 Shortcut-feature and explanation audit

The shortcut audit removes seven preregistered identity-, protocol-, window-, or behavior-related subsets from the frozen random-natural and chronological-natural populations across the frozen tree learners. Matched-size placebos, depth-one controls, feature-importance redistribution, behavior-restricted inputs, and family-composition tables test competing interpretations. PR-AUC and ROC-AUC are primary; operating-point results and conditional paired intervals retain their frozen roles. No subset is selected from target performance. Component TreeSHAP summaries are descriptive proxies for the averaged-probability ensemble. A deterministic local panel separately evaluates LIME decision agreement, surrogate fidelity, perturbation stability, and top-feature agreement with exact TreeSHAP. The audit establishes representation and split sensitivity; it cannot identify a feature as leakage or a cause of cross-domain failure.

4.4. 4.4 Cross-dataset transfer

Cross-dataset evaluation fits separate source models for IDS2018-to-CICIDS2017 and CICIDS2017-to-IDS2018 transfer. Bridge62 is primary; bridge70 is a preregistered serialization sensitivity. Source training, validation thresholds, feature contracts, and target cleaning are direction-specific and frozen independently of target results. Target labels do not guide fitting, feature selection, calibration, or thresholding. Direction-specific PR-AUC and ROC-AUC describe ranking; frozen-threshold metrics describe operating transfer; Brier score accompanies the aggregate-flag sensitivity. Published-versus-corrected paired intervals are conditional on the target and frozen

implementation. Cancelled `GROUNDED_S4` cells remain absent. The axis supports bridge- and direction-specific conclusions for two related benchmark families, not portability to arbitrary datasets or a unique transfer mechanism.

4.5. 4.5 Prevalence and operational stress

Operational stress translates inherited temporal and cross-dataset operating points without fitting or target access. Under prior-probability shift, sensitivity and specificity remain fixed while prevalence varies on a preregistered grid. PPV and negative predictive value follow from those rates and the assumed prior. Frozen traffic, analyst service time, capacity, and relative error weights produce alert, workload, capacity, and cost scenarios. These outputs are conditional projections. Their uncertainty is inherited from the operating points and assumptions; no new bootstrap is added. They show how a fixed operating point behaves under stated prevalence and resource scenarios, not how it would behave under simultaneous covariate, protocol, topology, user, or attacker shift.

4.6. 4.6 Deployment profiling

Profiling asks which frozen, compatible model paths are computationally feasible on the recorded hardware and software. The timed boundary begins with prepared model input and ends with materialized probabilities. Eligible tree, ensemble, and packet-image paths use frozen warm-inference, cold-start, throughput, memory, package-size, component, batch, and CPU/GPU schedules. Different representations remain separate, and unavailable, incompatible, timed-out, or resource-limited paths remain statuses rather than imputed timings. Latency percentiles, throughput, memory, and size retain native units; CPU/GPU ratios are descriptive point estimates. No batch is retrospectively chosen as optimal. The measurements exclude capture, flow extraction, external representation construction, alert aggregation, and analyst response and do not define hardware-independent constants.

4.7. 4.7 Leave-one-attack-family-out evaluation

LOAO evaluation asks whether XGBoost or LightGBM, with zero training and validation exposure to an eligible family, can rank that family above temporally matched benign traffic and detect it at frozen operating points. Family aliases, chronology, memberships, features, fits, and validation thresholds are fixed before target evaluation. DOS and AUTH_BRUTE_FORCE are ineligible; Infiltration is descriptive at support 36. ROC-AUC and PR-AUC

report ranking relative to chance and target prevalence. Recall and related metrics at standard, balanced, and security thresholds report operating transfer. Family-specific conditional intervals and support accompany the results, and behavioral-similarity analysis remains descriptive. This benchmark-family withholding design does not establish arbitrary future-attack detection or authorize a pooled novelty score.

4.8. 4.8 Seed and control stability

The stability program tests whether the central temporal ordering and eligible-family conclusions depend on one training realization. A frozen seed registry repeats designated temporal and LOAO cells without repeating the full historical hyperparameter search or selecting the best seed. Random-split LOAO is a control for chronology-related difficulty, not a deployment estimate. Seed-level PR-AUC, ROC-AUC, and operating outcomes are summarized separately from bootstrap uncertainty. Stability over five frozen seeds strengthens fixed-recipe conclusions within scope but supplies neither a population guarantee nor one causal account.

4.9. 4.9 Uncertainty, anti-adaptation, and reproducibility

Paired historical resampling keeps compared learners or representations on the same class-stratified draw. Each contrast retains its declared uncertainty type. An interval containing zero is not evidence of equivalence; an interval excluding zero does not broaden the population. Training-seed variation is not combined with bootstrap distributions, and unlike metrics are not normalized into a composite score. At each boundary, model families, hyperparameters, subsets, bridges, operating points, targets, and eligible families are frozen before the corresponding result. Later target evidence cannot retune an earlier protocol, and cancelled analyses remain cancelled. Manuscript claims and numbers map to frozen claim, evidence, number, source-artifact, limitation, and exhibit registries. The repository verifies source identities, configurations, schemas, scalar values, hashes, toy formulas, and approved read-only equivalence checks. It does not claim complete end-to-end reruns where raw data, historical environments, models, arrays, or closed targets are unavailable.

5. Results

5.1. 5.1 How strong was the conventional reference?

The processed reference condition produced strong within-table discrimination (Table 1). At the validation-selected balanced point, XGBoost achieved

holdout F1 0.9285, FPR 0.0061, ROC-AUC 0.9802, and PR-AUC 0.9776. The LightGBM security point achieved F1 0.9178, F2 0.9112, FPR 0.0466, ROC-AUC 0.9802, and PR-AUC 0.9777. These values define the reference against which the remaining validation questions are interpreted. Neither learner was declared the overall winner. The frozen balanced XGBoost-minus-LightGBM F1 difference was 0.000185, with a conditional interval from -0.000495 to 0.000862. At the security points, LightGBM missed 33 fewer attacks, with the paired conditional difference spanning 2 to 64 fewer misses. Brier scores were 0.04277 for XGBoost and 0.04275 for LightGBM, and the retained calibration-difference intervals included zero.

Table 1: Conventional processed-reference holdout results. Thresholds were selected on validation data. F1 and F2 are threshold-specific harmonic means; FPR is false-positive rate; ROC-AUC and PR-AUC are threshold-free ranking metrics. Dashes denote a metric not selected for this compact comparison.

Operating role	Learner	Threshold	F1	F2	FPR	ROC-AUC	PR-AUC
Balanced	XGBoost	0.51	0.9285	—	0.0061	0.9802	0.9776
Security	LightGBM	0.26	0.9178	0.9112	0.0466	0.9802	0.9777

5.2. 5.2 Did temporal development change forward validity?

On the common forward target, chronological-natural development ranked higher than random-natural development across all frozen seeds (Figure 1; Table 2). Random-natural models had mean ROC-AUC 0.5176 (SD 0.0073) and PR-AUC 0.2599 (SD 0.0034); chronological-natural models had mean ROC-AUC 0.8209 (SD 0.0085) and PR-AUC 0.6388 (SD 0.0322). Both metric orderings held in 5/5 seeds. Threshold transfer contradicted a simple ranking narrative. Chronological thresholds almost never fired on the common target, whereas the two random security points reached final FPR values of 0.2870 and 0.2685, far above the development constraint. Family-aware controls were mixed across family and learner, so the result does not isolate chronology as the sole mechanism.

5.3. 5.3 Were shortcut and explanation conclusions stable?

Shortcut-subset effects varied by development geometry and learner (Figure 2). The behavior-restricted representation retained strong random-split

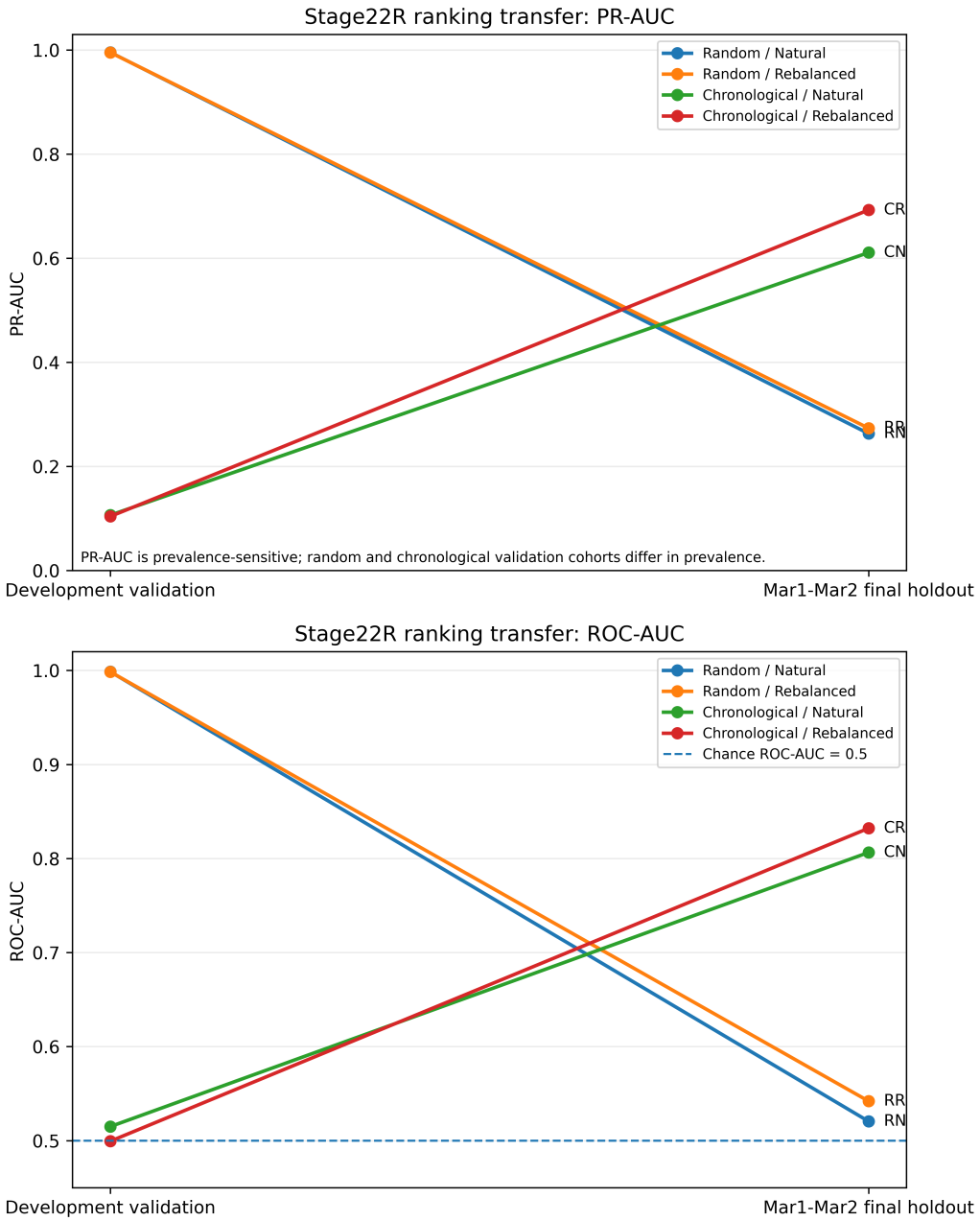


Figure 1: Temporal ranking transfer on the shared forward IDS2018 target. Existing panels report (a) PR-AUC and (b) ROC-AUC for frozen random and chronological development geometries. The five-seed stability summaries establish a 5/5 descriptive ordering, not a significance test; the target was historically opened, thresholds are evaluated separately, and chronology remains entangled with family composition.

Table 2: Five-seed temporal stability on the shared forward target. SD is training-seed standard deviation under the frozen recipe; the directional count is descriptive over the five registered seeds.

Development geometry	Mean ROC-AUC	ROC-AUC SD	Mean PR-AUC	PR-AUC SD	Directional result
Random-natural	0.5176	0.0073	0.2599	0.0034	Lower than chronological in 5/5 seeds for both metrics
Chronological-natural	0.8209	0.0085	0.6388	0.0322	Higher than random in 5/5 seeds for both metrics

discrimination but not the chronological result, and family composition was narrower in the chronological population. Matched-size placebo removals also produced non-zero split interactions, including frozen conditional intervals excluding zero. The observed interactions therefore establish feature-set sensitivity without identifying a named feature as leakage or a causal source of transfer behavior. Local explanations supplied a related counterexample. LIME rankings were comparatively stable across perturbation seeds, but only 2 of 64 explanations met all prespecified fidelity and cross-method criteria, and 31 of 64 failed to reproduce the model's local classification decision. Mean SHAP-LIME top-10 Jaccard agreement was 0.304 for XGBoost and 0.360 for LightGBM. Stable explanations were not necessarily faithful explanations.

5.4. 5.4 Was cross-dataset transfer symmetric?

Bridge62 transfer was strongly directional (Figure 3; Table 3). IDS2018-to-CICIDS2017 evaluation achieved PR-AUC 0.667483 and ROC-AUC 0.733946 on 2,830,743 effective target rows. In the reciprocal direction, target attack prevalence was 0.104847, PR-AUC was 0.108176, and ROC-AUC was 0.525167. The reverse PR-AUC was therefore close to its prevalence anchor, while the forward bridge retained substantial ranking signal. A larger bridge did not

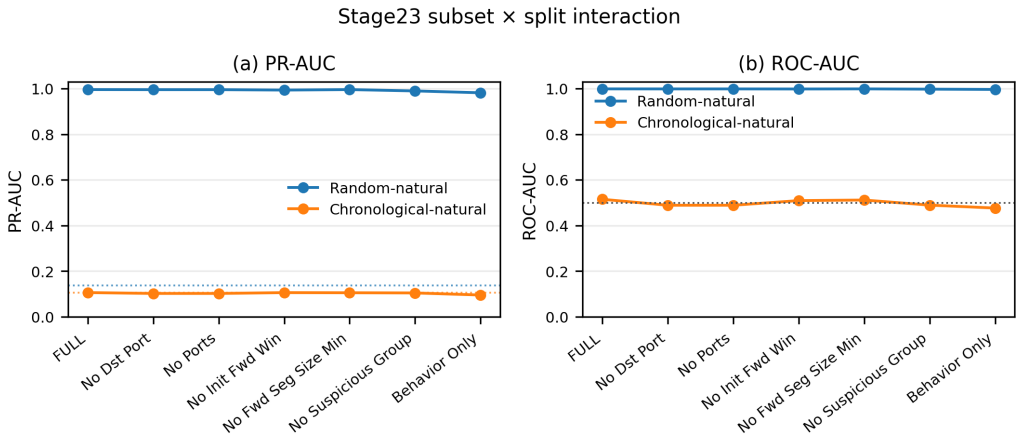


Figure 2: Shortcut-subset sensitivity by validation geometry. The approved asset reports native ranking behavior across seven frozen feature subsets, random-natural and chronological-natural splits, and the evaluated tree learners. Conditional subset effects do not identify leakage or a transfer mechanism; matched-size placebo interactions remain part of the interpretation.

yield one metric direction. Correcting aggregate-flag serialization in bridge70 changed PR-AUC by -0.007729, ROC-AUC by +0.002192, and Brier score by +0.005087; the frozen paired intervals excluded zero. The two `GROUNDED_S4` cells remained cancelled because exact membership was unavailable. Neither the metric-specific sensitivity nor the missing cells are hidden by averaging the directions.

Table 3: Primary bidirectional bridge62 transfer. PR-AUC is interpreted against each target's attack prevalence; ROC-AUC is the direction-specific ranking metric. Different source and target protocols preclude pooling.

Direction	Target scope	Target prevalence	PR-AUC	ROC-AUC
IDS2018 to CICIDS2017	2,830,743 effective rows	Target-specific	0.667483	0.733946
CICIDS2017 to IDS2018	Frozen February 28 target	0.104847	0.108176	0.525167

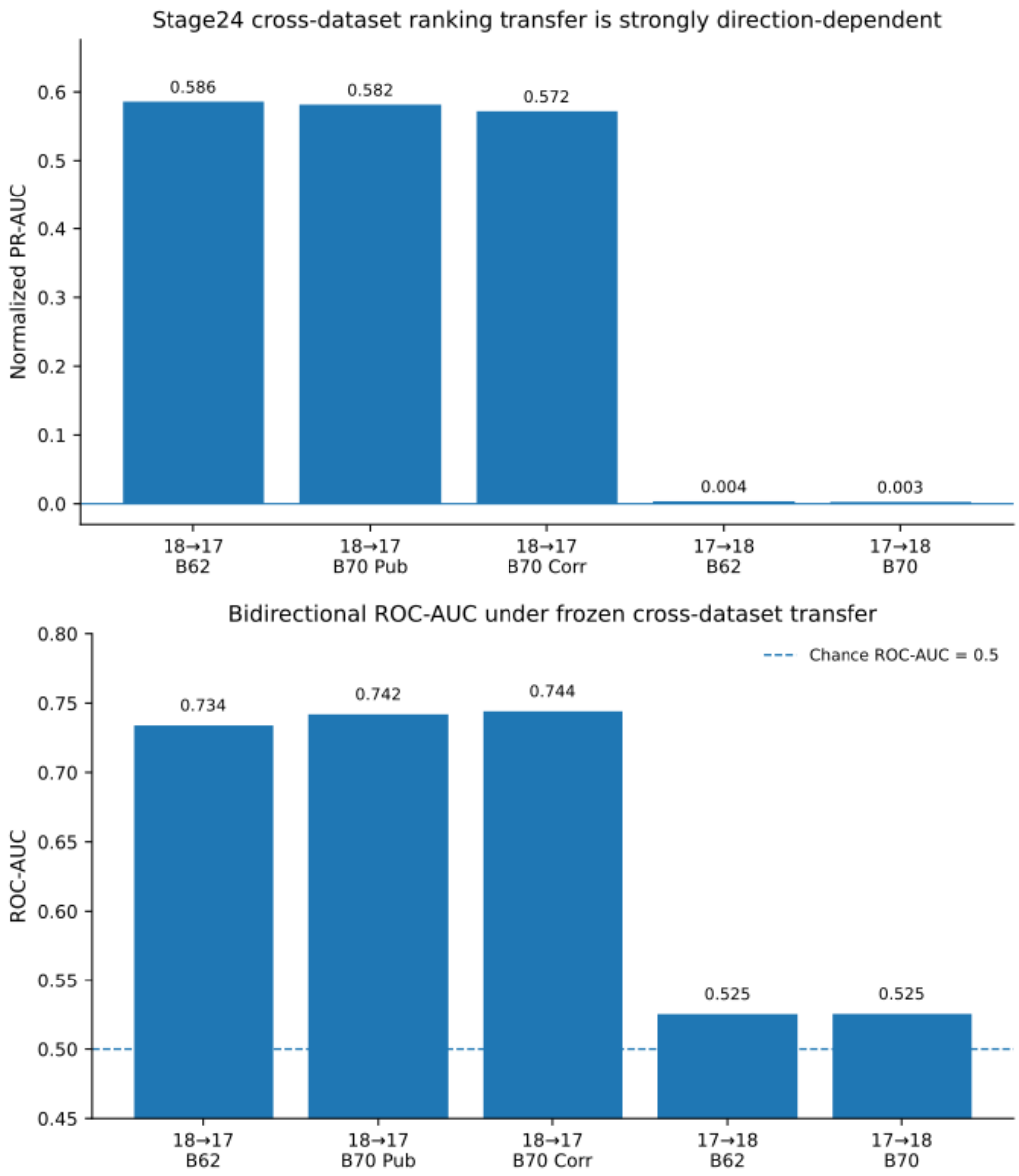


Figure 3: Bidirectional cross-dataset ranking under bridge62. Existing panels report normalized PR-AUC and ROC-AUC. Each point uses its frozen direction-specific source learner, target population, prevalence, and semantic feature contract; the panels do not define an average transfer score.

5.5. 5.5 Did projected precision imply operational utility?

Projected PPV, workload, and yield did not align uniformly (Figure 4; Table 4). At 0.1% assumed attack prevalence, the random-development STANDARD point retained PPV 0.965572 but required 33.5 analyst-hours per day under the frozen traffic and service-time scenario. The chronological-development STANDARD point projected PPV 0.000551068 and only 0.0322 true alerts per day: its small workload reflected negligible yield. Directionality persisted under the same prior-shift translation. Forward STANDARD transfer points projected PPV 0.039233-0.060313, while reverse points projected 0.000257610-0.000287993. Under the frozen 1:100 relative-cost scenario, 15/24 operating points favored the model at 0.1% prevalence and 3/24 at 0.01%. These are analytic projections with fixed sensitivity and FPR, not observations from a security operations center.

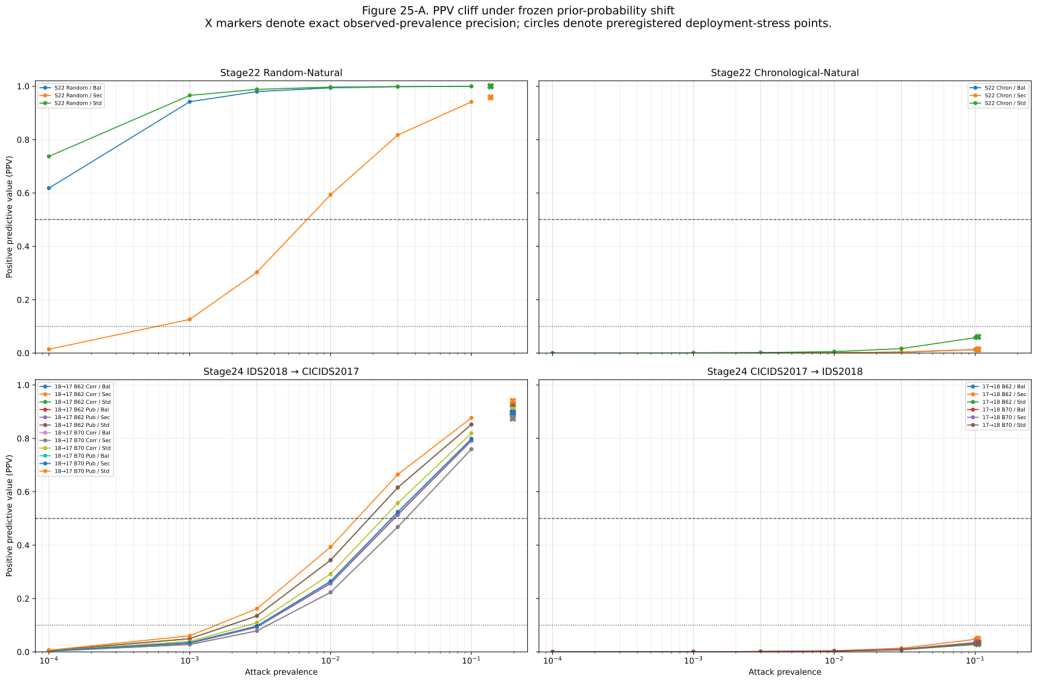


Figure 4: PPV under prevalence stress. The approved curves translate frozen true- and false-positive rates across assumed attack prevalence. PPV is positive predictive value. The prior-shift model holds conditional rates fixed; traffic, service time, capacity, and relative costs are scenario assumptions rather than field measurements.

Table 4: Selected operating-point translations at 0.1% assumed prevalence. PPV and workload/yield are conditional on the frozen rates, daily traffic, and analyst service-time assumptions; STANDARD denotes the inherited standard operating role.

Frozen condition	Projected PPV	Operational quantity	Interpretation boundary
Random-development STANDARD	0.965572	33.5 analyst-hours/day	High PPV does not ensure manageable workload
Chronological-development STANDARD	0.000551068	0.0322 true alerts/day	Low workload does not ensure useful yield
Forward transfer STANDARD range	0.039233-0.060313	Direction-specific	Conditional prior-shift projection
Reverse transfer STANDARD range	0.000257610-0.000287993	Direction-specific	Conditional prior-shift projection

5.6. 5.6 Was one compute backend consistently faster?

Backend advantage depended on architecture and measurement path (Figure 5). At batch one, the p95 CPU-over-GPU latency ratio was 1.94 for the five-checkpoint soft-voting ensemble and 1.58 for the single-resource reference, indicating a GPU advantage. Ratios of 0.16 for CatBoost and 0.26 for XGBoost indicated a CPU advantage under the same convention. The packet-image CNN ratio was 10.86, while the ViT ratio was 1.05. Unsupported LightGBM GPU execution and unavailable paths remained compatibility statuses rather than timing values.

5.7. 5.7 Did eligible withheld families behave uniformly?

LOAO results were selective (Figure 6; Table 5). DDoS ranking was strong for XGBoost (ROC-AUC 0.9982; PR-AUC 0.9925) and LightGBM (ROC-AUC 0.9986; PR-AUC 0.9940). Web Attack also retained ranking for XGBoost (ROC-AUC 0.9693; PR-AUC 0.7206) and LightGBM (ROC-AUC 0.9901; PR-AUC 0.7605). Other families were more conditional. Bot produced XGBoost ROC-AUC 0.3224 and PR-AUC 0.003256 but LightGBM ROC-AUC 0.5591. Port Scan ROC-AUC was 0.5506 for XGBoost and 0.7559 for LightGBM. Infiltration remained descriptive at 36 positives, and DOS and

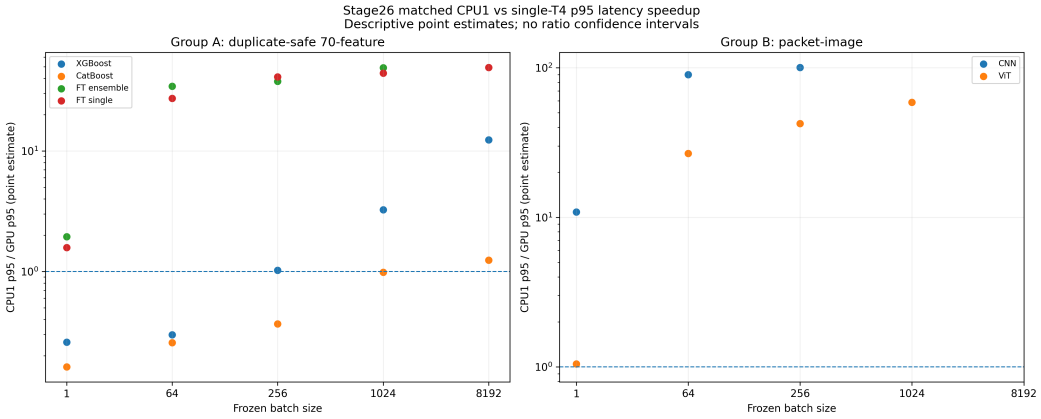


Figure 5: Hardware-specific p95 CPU/GPU comparison. The approved asset compares batch-one p95 latency ratios for compatible prepared-input-to-probability paths on the recorded hardware and software. Values are descriptive point estimates; unsupported backends, capture, flow extraction, alert aggregation, and analyst response are outside the comparison.

AUTH_BRUTE_FORCE were ineligible rather than model failures. Ranking and frozen-threshold recall could also disagree, so no pooled novelty score was formed.

5.8. 5.8 Did central conclusions survive the frozen realizations?

All 108/108 preregistered new fits in the frozen stability program completed, with 12 artifacts reused as declared and no planned execution outstanding. Completion did not enlarge inferential scope: no best seed was selected, seed variation was not combined with bootstrap uncertainty, and the shared temporal target remained historically opened. The chronological-over-random ranking direction held in all evaluated seeds. DDoS, Port Scan, and Web Attack met the registered qualitative conditions for both learners; Bot remained learner-dependent and Infiltration descriptive. Random LOAO controls were mixed and family-specific. The controls support an interaction between novelty difficulty and chronology in some comparisons but do not identify one mechanism.

5.9. 5.9 What survived across the validation axes?

Table 6 synthesizes the frozen chain without treating unlike quantities as commensurable. Strong benchmark discrimination, chronological forward

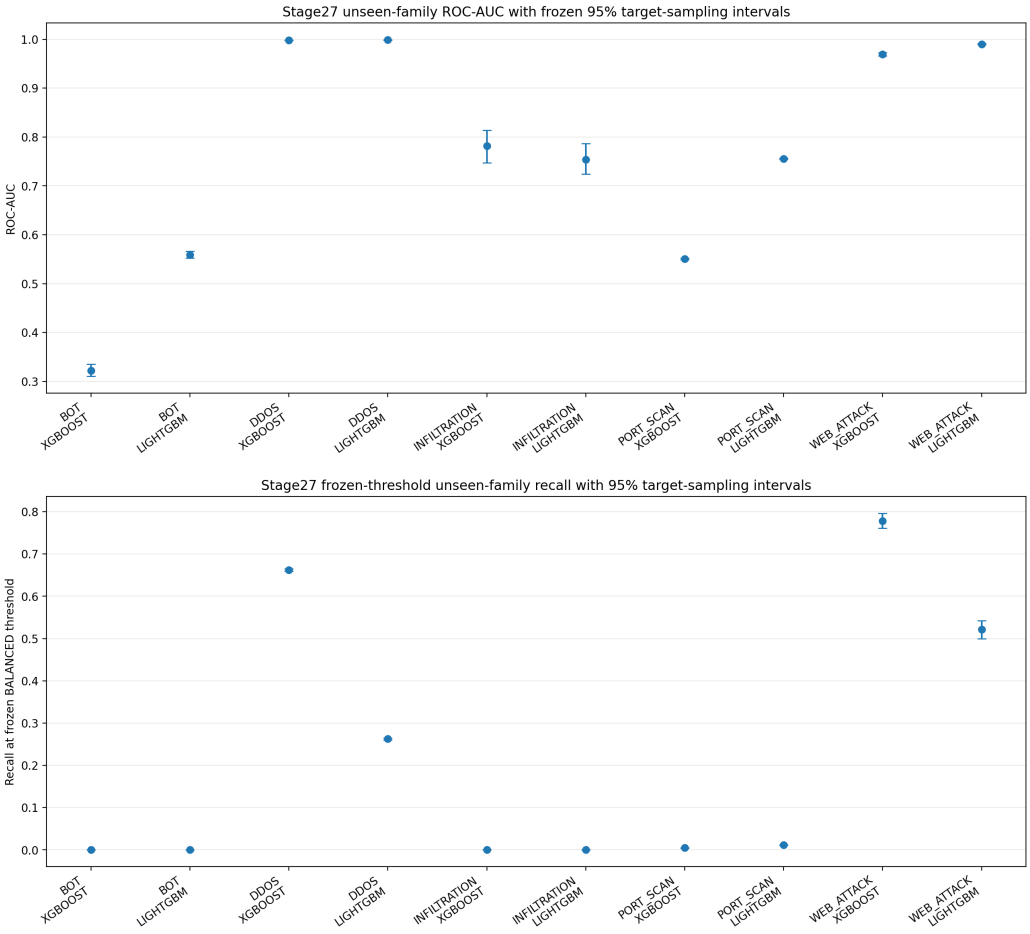


Figure 6: Eligible-family ranking and balanced-threshold recall. Existing panels report ROC-AUC with frozen conditional intervals and balanced-threshold recall for separate eligible families and learners. Infiltration is descriptive at support 36; ineligible families are absent; the panels do not constitute an aggregate future-attack score.

Table 5: Frozen conclusion-stability classification for eligible families. Classifications combine family-specific ranking, frozen-threshold behavior, support, learner contrast, and the five-seed stability rules; they are not pooled metrics.

Family	Frozen synthesis	Required qualification
DDoS	Strong ranking for both learners	Eligible benchmark family only
Web Attack	Ranking retained for both learners	Magnitude and thresholds remain learner-specific
Port Scan	Qualitative conditions retained across seeds	Ranking magnitude is learner-dependent
Bot	Learner-dependent	Frozen-threshold detection remains weak
Infiltration	Descriptive only	Held-out positive support is 36

ranking, forward bridge signal, selected family transfer, and compatible computational paths survived within their declared scopes. Frozen thresholds, symmetric portability, simple shortcut causality, universal operational preference, and uniform family conclusions did not.

6. Discussion

6.1. 6.1 Stronger validation changed the claim, not every result

The reference experiment remains a credible answer to a narrow question: selected tree ensembles discriminated the processed holdout labels well under frozen validation choices. Stronger validation did not erase that arithmetic or make benchmark evaluation uninformative. Instead, each axis changed the population, operating rule, representation, or resource assumption to test a broader inference. The resulting pattern was not monotonic. Chronological development retained substantially stronger forward ranking than random development. The forward semantic bridge retained cross-dataset signal, DDoS and Web Attack remained discriminable when withheld, and compatible model paths were feasible on the measured system. Other findings were less favorable: frozen thresholds transferred poorly, reverse cross-dataset ranking was near its anchor, and some family outcomes depended on learner or support. Benchmark performance is therefore a property of one evaluation system. A

Table 6: Integrated native-metric validity matrix. Each row preserves its population, metric family, and interpretation ceiling. The table summarizes frozen evidence and is not a normalized degradation, generalization, or composite performance score.

Validity axis	Principal observation	What survived	What remained conditional or unsupported	Claim ceiling
Benchmark/reference	Strong F1 and ranking on the processed split	Within-reference discrimination	Inference to chronology, transfer, or deployment	Processed reference only
Duplicate/split	Frozen memberships and target roles were auditable	Explicit role separation	Latent-session independence	Traceable memberships only
Temporal	Chronological ranking exceeded random ranking across five seeds	Directional ranking contrast	Threshold transfer and sole mechanism	Evaluated chronology and recipe
Shortcut	Subset effects depended on split and learner; placebos interacted	Representation sensitivity	Universal removal benefit or causal leakage	Preregistered subsets and controls
Cross-dataset	Forward ranking signal contrasted with near-baseline reverse transfer	Direction-specific forward signal	Symmetry, arbitrary datasets, cancelled cells	Frozen bridges and targets
Prevalence/operational	PPV, workload, yield, and cost preference diverged	Assumption-explicit translation	Field performance or universal preference	Frozen prior-shift scenarios
Computational	Backend advantage varied by compatible model path	Measured component feasibility	Unsupported paths and end-to-end cost	Recorded hardware/software
Unseen family	Several families transferred; others depended on learner or support	Selected eligible-family capability	Uniform future-attack detection	Eligible benchmark families
Seed/control	Central directions survived selected seeds; controls were mixed	Finite-realization stability	Population guarantee or one causal account	Frozen seed registry

broader capability claim must name the additional condition under which it was validated.

6.2. 6.2 Temporal, shortcut, and explanation evidence constrain mechanism claims

The temporal evidence separates ranking from score-scale transfer. Chronological development ranked the shared forward population more effectively in every frozen realization, yet its inherited thresholds almost never fired. This does not show that chronological development is universally preferable. It shows that validation geometry can change both ranking and operating behavior. Family-aware controls remain mixed, so chronology or drift alone cannot explain the contrast. Shortcut controls reinforce that restraint. Named removals altered results, but matched placebos also interacted with split geometry. Chronological family composition also differed from random composition. The study can report subset sensitivity without treating a field as proven leakage or as the cause of domain transfer. LIME provides a related warning: perturbation rankings could be stable while local decision reconstruction and cross-method fidelity failed. Explanation reports should measure fidelity and agreement rather than equate repeatability with validity.

6.3. 6.3 Transfer is directional and threshold utility is a separate decision

The bidirectional bridge makes dataset portability a source-to-target statement. IDS2018-trained models retained useful ranking on CICIDS2017 under bridge62, while the reciprocal evaluation was close to its target anchor. The directions use different source learners, prevalence, collection periods, and target roles. Averaging them would remove the finding. Metric-specific responses to the aggregate-flag correction also show that a larger feature intersection is not automatically a better semantic bridge. The cancelled cells preserve another boundary: exact membership governance can leave a planned comparison unestimable. Ranking and operating utility also diverged across temporal, cross-dataset, and LOAO results. ROC-AUC and PR-AUC integrate score order across thresholds; precision, recall, F1, and FPR describe one selected point. A transported ranking can coexist with an unsuitable threshold when score scale or prevalence changes. Prospective recalibration might be useful, but it would require new governed data and evaluation. Target-guided recalibration was not performed here.

6.4. 6.4 Operational projections and compute profiles answer different questions

Prior-shift translation demonstrates that PPV alone does not determine workflow. Under the frozen scenarios, high projected PPV coexisted with analyst demand exceeding one day, while low demand could result from almost no true detections. Relative-cost preference also changed with assumed prevalence. Evaluation protocols should report the inherited rates and the assumed prevalence, traffic, service time, capacity, and error costs. An operating point should not be labeled operationally useful from PPV alone. The profiling analysis asks a complementary systems question: can a compatible prepared-input-to-probability path run in the declared environment? CPU/GPU advantage varied by model family, and unsupported paths remained statuses. These measurements neither validate PPV nor cover capture-to-alert latency. Operational analysis and computational profiling belong in the same decision chain, but their evidence boundaries remain separate.

6.5. 6.5 Withheld-family evidence must remain heterogeneous

Zero exposure during training and validation did not produce one family outcome. DDoS and Web Attack transferred strongly across learners. Port Scan retained qualitative stability but differed in magnitude, and Bot was

learner-dependent. Infiltration remained descriptive, while two families were ineligible. One novelty score would hide these differences in support, threshold behavior, and learner dependence. This heterogeneity is informative. Strong family results prevent a universal negative conclusion. Weak and ineligible cases prevent a universal positive one. A withheld benchmark family can share infrastructure, extraction cues, or behavior with known classes; it is not equivalent to an arbitrary future exploit. Future evaluations should preregister eligibility, retain family support, report ranking and frozen-threshold behavior separately, and distinguish descriptive from inferential outcomes.

6.6. 6.6 An Eight-Item Validation Checklist

The evidence supports eight claim checks. Each check uses its own metrics and controls; completing the list is not a universal guarantee of deployment validity.

1. **Define the reference population** Report sampling, prevalence, duplicate handling, and split construction. Use membership counts and class support as the native evidence, and state which population the resulting metrics describe.
2. **Test temporal validity** Use forward or chronological separation when it matches the intended claim. Report ranking metrics and operating-point metrics separately, and retain controls for changes in family composition or other competing mechanisms.
3. **Probe shortcut sensitivity** Ablate suspicious features and, where appropriate, compare behavior-restricted representations. Pair named interventions with negative or matched-placebo controls; an effect establishes sensitivity, not leakage or causality.
4. **Evaluate cross-dataset transfer bidirectionally** Declare the semantic feature bridge and report each source-to-target direction separately. Prohibit target adaptation or identify it explicitly, and do not average asymmetric transfer away.
5. **Stress prevalence assumptions** State the target prevalence and translate frozen rates into PPV, alert volume, and workload. Distinguish an analytic prior-shift projection from measured field performance and disclose traffic, service-time, capacity, and cost assumptions.
6. **Profile the declared deployment boundary** Report latency or throughput with the hardware, software, batch, and timed component. Include feature-extraction cost when measured; otherwise state that it is outside the boundary, along with unsupported runtime paths.

7. **Test unseen-family behavior with eligibility controls** Verify zero training and validation exposure to the held-out family, publish eligibility criteria and support, and report ranking and frozen-threshold behavior separately. Keep low-support families descriptive.
8. **Test realization stability** Repeat critical conclusions across registered seeds or model realizations without selecting the best run. Report realization spread separately from bootstrap or target-sampling uncertainty.

These checks are modular. The appropriate subset depends on the claim, population, and intended use, and no item substitutes for another validity axis. Claim-to-artifact links should record the frozen decision, evidence source, and interpretation ceiling for each completed check.

7. Limitations

The governed limitations fall into six conceptual groups. The complete limitation-to-claim-ceiling matrix is provided with the supplementary and repository materials.

7.1. 7.1 Dataset and benchmark scope

The evidence covers two related benchmark families, CSE-CIC-IDS2018 and CICIDS2017, rather than current operational networks (`LIM29-001`). The processed reference is rebalanced and stratified, so its high metrics do not establish natural-prevalence, session, temporal, or deployment validity (`LIM29-002`). Some raw data and historical scientific artifacts are unavailable or intentionally closed (`LIM29-015`). The graph result is restricted to one authenticated source and cannot establish unseen-host or full-reference capability (`LIM29-016`). The CNN/ViT comparison is a single descriptive contrast, not evidence of general architecture superiority (`LIM29-017`).

7.2. 7.2 Temporal and family-composition confounding

Chronological memberships also change attack-family composition. Mixed controls therefore prevent attribution of the temporal contrast to chronology or concept drift alone (`LIM29-003`). The stability analysis reuses the historically opened shared target; it strengthens the frozen conclusion across realizations but is not an independent blind replication (`LIM29-018`).

7.3. 7.3 Cross-dataset and family-support constraints

Cross-dataset conclusions depend on bridge62/70 semantics, preprocessing, learners, direction, and target roles; they do not establish bridge-independent or symmetric portability (`LIM29-004`). Two `GROUNDED_S4` cells were cancelled because exact membership was unrecoverable, so not every planned sensitivity was executed (`LIM29-005`). LOAO coverage is also incomplete: DOS and AUTH_BRUTE_FORCE were structurally ineligible (`LIM29-006`), and Infiltration remains descriptive at 36 held-out positives (`LIM29-007`). These constraints preclude uniform conclusions across datasets or attack families.

7.4. 7.4 Statistical and realization uncertainty

Paired confidence intervals are conditional on the frozen target samples, fitted models or arrays, and implementations; they do not describe unconditional deployment uncertainty (`LIM29-008`). Five registered seeds assess a fixed recipe without repeating full hyperparameter optimization. The 5/5 ordering is descriptive, not a significance test or population guarantee (`LIM29-009`).

7.5. 7.5 Operational and hardware assumptions

The prior-shift analysis holds true- and false-positive rates fixed while prevalence changes. It measures sensitivity to that assumption, not performance under simultaneous covariate, protocol, topology, user, or attacker shift (`LIM29-012`). Traffic, analyst time, capacity, and relative costs are scenarios rather than organization-specific measurements (`LIM29-013`). Profiling covers compatible inference components on one hardware/software environment and omits complete capture-to-alert processing (`LIM29-011`). It cannot establish universal workload, economic value, latency, throughput, energy, or end-to-end cost.

7.6. 7.6 Historical reproducibility

Some dependencies and execution environments remain `VERSION_NOT_PROVEN`, so bit-for-bit reconstruction of every historical environment is unsupported (`LIM29-010`). Static provenance, scalar evidence, configurations, archives, and equivalence tests make the manuscript chain auditable, but missing data and closed artifacts prevent some end-to-end reruns (`LIM29-014`). The repository therefore supports claim-level traceability, not complete local regeneration of every result.

8. Conclusion

A processed benchmark score is a necessary reference, not a complete capability claim. Strong reference discrimination survived the evaluated validation chain. Chronological development retained higher forward ranking across five seeds, the forward cross-dataset bridge retained signal, and several eligible withheld families remained discriminable. Compatible inference paths were also feasible on the recorded hardware. Other conclusions remained conditional. Frozen thresholds transferred poorly even when ranking survived. Reverse cross-dataset transfer was near its anchor. Shortcut and placebo interactions did not isolate one mechanism, operational projections depended on prevalence and workload assumptions, and withheld-family behavior varied by learner and support. Future evaluations should name the validity condition, population, metric, operating rule, uncertainty, and adaptation boundary attached to each claim. Benchmark, temporal, shortcut, domain, operational, computational, unknown-family, and realization evidence should remain separate. This preserves failures as well as non-collapse findings. The resulting account states which capability was observed, where it survived, and what the evidence does not establish. Stronger validation narrows or qualifies a claim; it does not require a monotonic performance decline. It also does not justify a single failure label or a universal deployment verdict.

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